

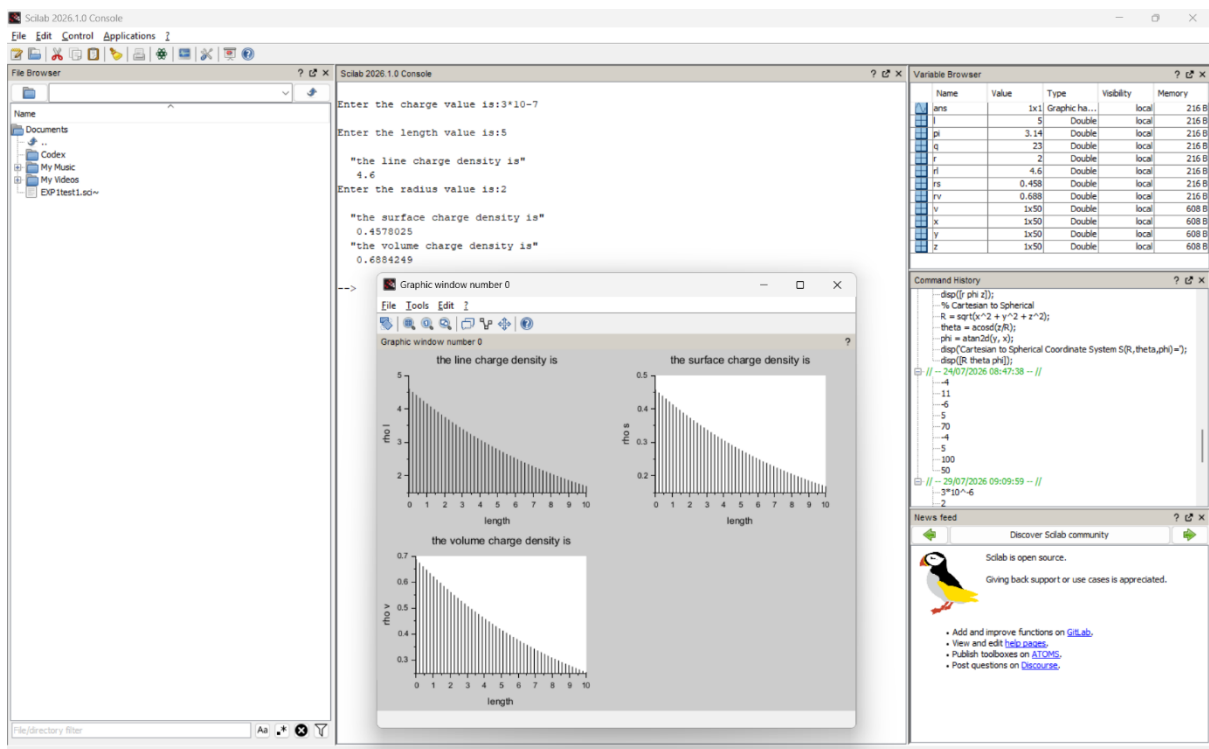
EXP 4: Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

```

1 clear;
2 clear;
3 q=input('Enter the charge value is:');
4 l=input('Enter the length value is:');
5 //line charge density
6 z1=q/l;
7 disp('the line charge density is',[z1]);
8 //surface charge density
9 pi=3.14;
10 r=input('Enter the radius value is:');
11 ra=q/(4*pi*r^2);
12 disp('the surface charge density is',[ra]);
13 //volume charge density
14 rv=q/(4.33*pi*r^3);
15 disp('the volume charge density is',[rv]);
16 x = linspace(0, 10, 50);
17 y = z1*exp(-0.1 * x);
18 z = ra*exp(-0.1 * x);
19 v = rv*exp(-0.1 * x);
20 figure();
21 subplot(3, 2, 1);
22 plot2d3(x,y);
23 title('the line charge density is');
24 xlabel('length');
25 ylabel('rho l');
26 subplot(3, 2, 2);
27 plot2d3(x,z);
28 title('the surface charge density is');
29 xlabel('length');
30 ylabel('rho a');
31 subplot(3, 2, 3);
32 plot2d3(x,v);
33 title('the volume charge density is');
34 xlabel('length');
35 ylabel('rho v');

```

Line 35, Column 16.



exp 4 :-

$$\text{charge} = 3 \times 10^{-7}$$

$$\text{length} = 5 \text{ cm}$$

$$\text{radius} = 2 \text{ mm}$$

line charge density :- $\frac{q}{l}$

$$= \frac{23}{5} = 4.6$$

$$\boxed{\text{Line charge density} = 4.6}$$

surface density

$$\rho_s = \frac{q}{4\pi r^2}$$

$$= \frac{23}{4 \times 3.14 \times 2^2}$$

$$= \frac{23}{50.24}$$

$$\boxed{\rho_s = 0.4578025}$$

volume charge density

$$\rho_v = \frac{q}{1.33\pi r^3}$$

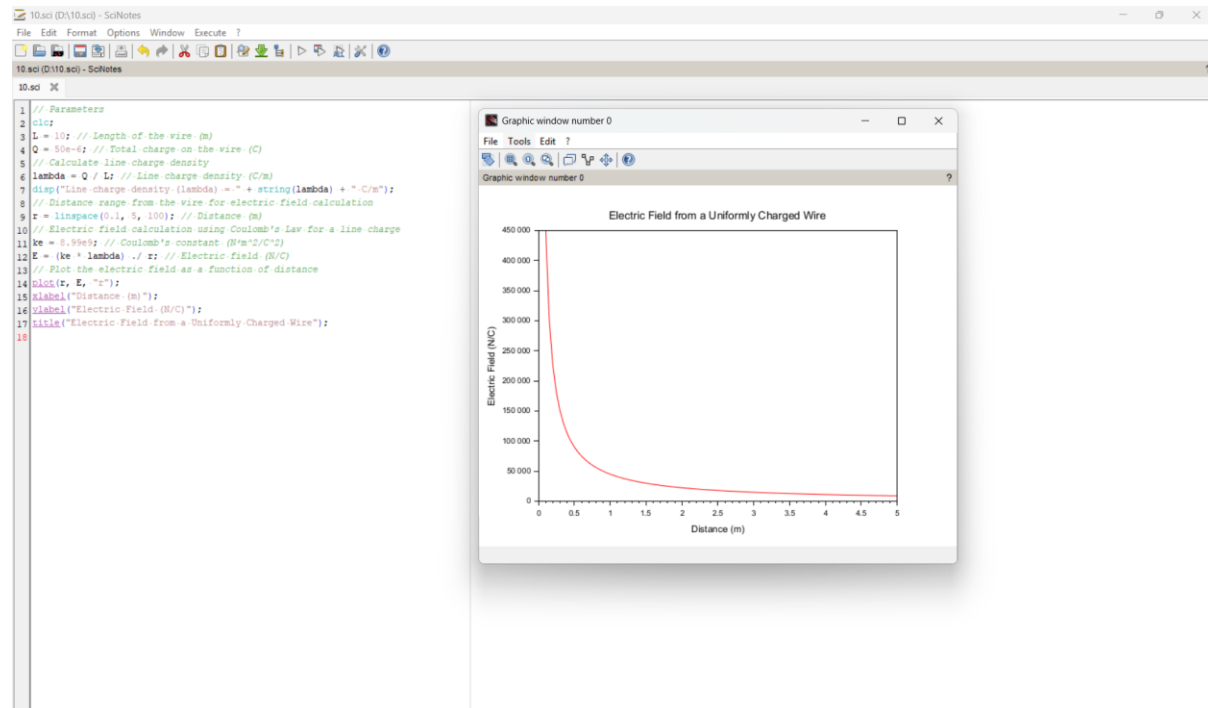
$$= \frac{23}{1.33 \times 3.14 \times 2^3}$$

$$= \frac{23}{33.4096}$$

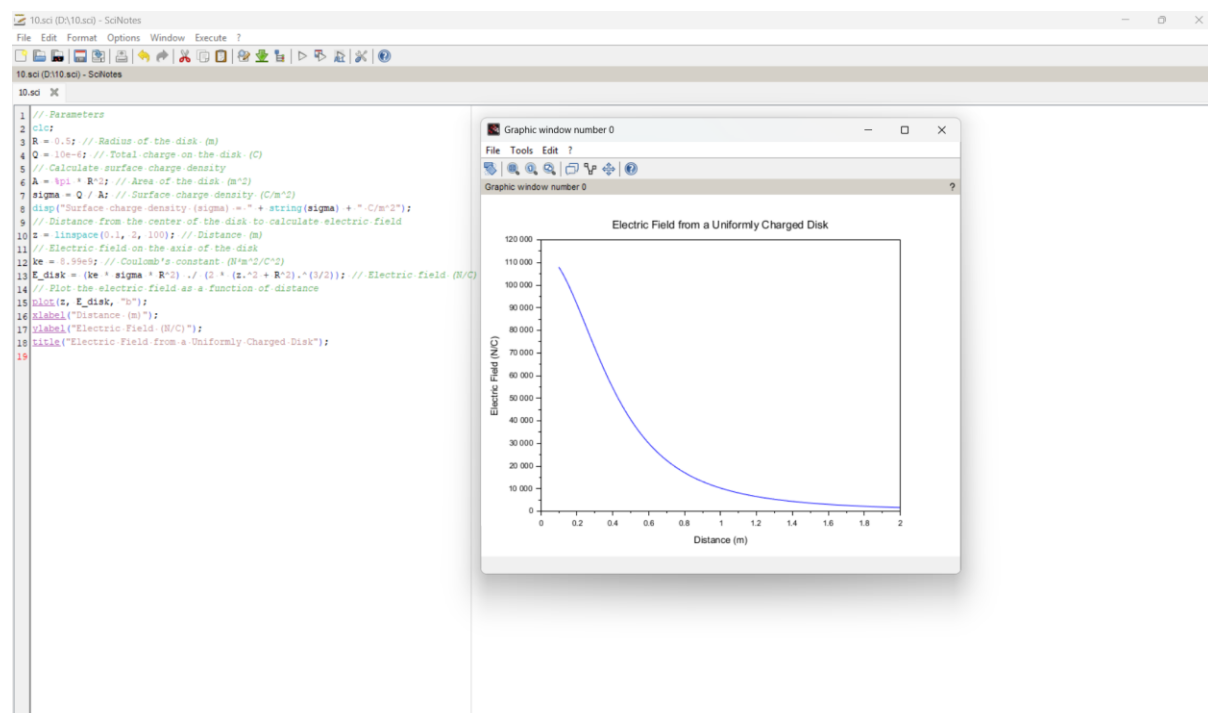
$$\boxed{\rho_v = 0.6884249}$$

Case 1:

A long wire of length $L=10\text{m}$ carries a uniform charge of $Q=50\mu\text{C}$. The goal is to calculate the line charge density λ and determine how the electric field behaves as a function of distance from the wire.



Case 2: Given a uniformly charged disk of radius $R=0.5\text{m}$ with total charge $Q=10\mu\text{C}$, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behavior of the electric field as a function of distance from the center of the disk.



Case 3: Given a uniformly charged sphere of radius $R=2\text{m}$ with total charge $Q=20\mu\text{C}$, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.

